



Polar Coordinates · Mark Scheme

Core Pure 2 Chapter 5: Polar Coordinates. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Polar and Cartesian forms

- 1 $r = \sqrt{9+27} = 6$ B1; point in the second quadrant, $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$, so $(6, \frac{2\pi}{3})$ B1.
- 2 $x = 4 \cos(-\frac{3\pi}{4})$, $y = 4 \sin(-\frac{3\pi}{4})$ M1; $(-2\sqrt{2}, -2\sqrt{2})$ A1.
- 3 $(r^2)^2 = 4(r \cos \theta)(r \sin \theta)$ M1; $r^4 = 4r^2 \sin \theta \cos \theta = 2r^2 \sin 2\theta$ M1; $r^2 = 2 \sin 2\theta$ A1.
- 4 $r - r \cos \theta = 2$ M1; $r = 2 + x$ M1; $x^2 + y^2 = (2+x)^2$ M1; $y^2 = 4x + 4$ A1.
- 5(a) $r^2 = 4r \cos \theta \Rightarrow x^2 + y^2 = 4x$ M1 A1; $(x-2)^2 + y^2 = 4$: circle, centre (2,0), radius 2 A1.
- 5(b) $r \cos \theta = 3$ M1; $r = 3 \sec \theta$ A1.
- 5(c) $4 \cos \theta = 3 \sec \theta \Rightarrow \cos^2 \theta = \frac{3}{4}$ M1; $\theta = \pm \frac{\pi}{6}$ A1; $(2\sqrt{3}, \frac{\pi}{6})$ and $(2\sqrt{3}, -\frac{\pi}{6})$ A1.

Section B – Sketching polar curves

- 6 Circle centre O , radius 3 B1; half-line from O at $\frac{3\pi}{4}$ to the initial line (into the second quadrant), not a full line B1.
- 7 Spiral starting at O , r increasing anticlockwise, one full turn B1; meets $\theta = \pi$ at $(2\pi, \pi)$ B1; ends on the initial line at $(4\pi, 2\pi)$, i.e. the point $(4\pi, 0)$ B1.
- 8 Closed convex (egg-shaped) curve around O , symmetric about the initial line B1; $(5, 0)$ and $(3, \pi)$ B1; $(4, \pm \frac{\pi}{2})$ B1.
- 9 (a) $\sin 3\theta \geq 0$: $-\frac{2\pi}{3} \leq \theta \leq -\frac{\pi}{3}$, $0 \leq \theta \leq \frac{\pi}{3}$, $\frac{2\pi}{3} \leq \theta \leq \pi$ B1. (b) Three equal loops meeting at O B1; tips at $(2, \frac{\pi}{6})$ and $(2, \frac{5\pi}{6})$ B1; and $(2, -\frac{\pi}{2})$ B1.
- 10(a) Cardioid symmetric about the initial line, with a cusp at O B1; passes through $(4, 0)$ B1.
- 10(b) $y = r \sin \theta = 2 \sin \theta + 2 \sin \theta \cos \theta = 2 \sin \theta + \sin 2\theta$ M1; $\frac{dy}{d\theta} = 2 \cos \theta + 2 \cos 2\theta = 0$ M1; $2 \cos^2 \theta + \cos \theta - 1 = (2 \cos \theta - 1)(\cos \theta + 1) = 0$, $\cos \theta = \frac{1}{2}$ ($\cos \theta = -1$ gives the pole) M1; $(3, \frac{\pi}{3})$ and $(3, -\frac{\pi}{3})$ A1.
- 10(c) $x = r \cos \theta = 2 \cos \theta + 2 \cos^2 \theta$, $\frac{dx}{d\theta} = -2 \sin \theta - 4 \sin \theta \cos \theta$ M1; $-2 \sin \theta (1 + 2 \cos \theta) = 0 \Rightarrow \theta = 0$ or $\cos \theta = -\frac{1}{2}$ M1; $(4, 0)$, $(1, \frac{2\pi}{3})$, $(1, -\frac{2\pi}{3})$ A1.

Section C – Area enclosed by a polar curve

- 11 $\frac{1}{2} \int_0^\pi 4\theta^2 d\theta = [\frac{2}{3}\theta^3]_0^\pi$ M1; $= \frac{2\pi^3}{3}$ A1.
- 12 One loop: $\frac{1}{2} \int_0^{\pi/3} 4 \sin^2 3\theta d\theta = \int_0^{\pi/3} (1 - \cos 6\theta) d\theta$ M1; $[\theta - \frac{1}{6} \sin 6\theta]_0^{\pi/3}$ A1; $= \frac{\pi}{3}$ A1.
- 13 $\frac{1}{2} \int_{-\pi}^\pi (16 + 8 \cos \theta + \cos^2 \theta) d\theta$ M1; $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$, giving $\frac{1}{2} [\frac{33}{2}\theta + 8 \sin \theta + \frac{1}{4} \sin 2\theta]_{-\pi}^\pi$ M1; $= \frac{33\pi}{2}$ A1.
- 14 $\frac{1}{2} \int_0^{\pi/4} (1 + 2 \tan \theta + \tan^2 \theta) d\theta$ M1; $= \frac{1}{2} \int_0^{\pi/4} (\sec^2 \theta + 2 \tan \theta) d\theta$ M1;
 $= \frac{1}{2} [\tan \theta + 2 \ln \sec \theta]_0^{\pi/4}$ M1; $= \frac{1}{2} (1 + 2 \ln \sqrt{2}) = \frac{1}{2} (1 + \ln 2)$ A1.
- 15(a) $3 \cos \theta = 1 + \cos \theta \Rightarrow \cos \theta = \frac{1}{2}$ M1; $\theta = \pm \frac{\pi}{3}$, $r = \frac{3}{2}$ A1.
- 15(b) By symmetry, area $= 2 \times \frac{1}{2} \int_0^{\pi/3} (9 \cos^2 \theta - (1 + \cos \theta)^2) d\theta$ M1;
 $= \int_0^{\pi/3} (8 \cos^2 \theta - 2 \cos \theta - 1) d\theta$ M1; $= \int_0^{\pi/3} (3 + 4 \cos 2\theta - 2 \cos \theta) d\theta$ M1; $= [3\theta + 2 \sin 2\theta - 2 \sin \theta]_0^{\pi/3}$ M1 A1; $= \pi +$



$$\sqrt{3} - \sqrt{3} = \pi \text{ A1.}$$

Roar-level challenge

16 $2 \sin 2\theta = \sqrt{2} \Rightarrow \sin 2\theta = \frac{1}{\sqrt{2}} \text{ M1; } \theta = \frac{\pi}{8}, \frac{3\pi}{8} \text{ A1; area} = \frac{1}{2} \int_{\pi/8}^{3\pi/8} (4 \sin^2 2\theta - 2) d\theta \text{ M1; } = \frac{1}{2} \int_{\pi/8}^{3\pi/8} (-2 \cos 4\theta) d\theta =$
 $\left[-\frac{1}{4} \sin 4\theta\right]_{\pi/8}^{3\pi/8} \text{ M1; } = -\frac{1}{4}(-1 - 1) = \frac{1}{2} \text{ A1.}$